

# TRIAD: Finite Complete-Plan Selection for Predictive Human-to-Robot Handover

Event time, grasp and route chosen together, certified through retreat, committed once

Hamidreza (Harry) J. Azari, Mourad Benoussaad, Wael Suleiman, Wiem Bel Hedi  
Laboratoire Génie de Production (LGP), ENIT, Université de Toulouse, Tarbes, France

**Abstract**—A robot that receives an object from a moving human hand must decide when the exchange will take place, how it will hold the object and along which path it will get there. TRIAD is a receiver-side controller for a Kinova Gen3 arm with a Robotiq 2F-85 gripper, implemented on `mc_rtc`, that makes this decision explicitly. From one frozen planning state it predicts the object’s presentation pose at 14 candidate event times, generates 32 grasps about the handle axis and 17 transit routes per grasp, and previews every complete plan of the resulting bank with a whole-arm kinematic rollout through reach, insertion, closure and carried-object retreat. Surviving plans are ranked by a seven-term objective referred to a common temporal origin, timing admission is re-applied with the controller clock at the instant of selection, and exactly one plan is committed and executed by the task-space QP layer. The search runs on a background worker so that the 1 kHz control thread never blocks. The paper gives the construction of each bank, the certification checks, the objective and the admission rule, and shows on the repository’s own records how a bank of 7616 candidates reduces to a few hundred certified plans, between ten and a few hundred admissible ones and a single commitment. The evidence comprises four reference scenarios, a perception-latency sweep in which delay compensation extends completion from 0.22 s to 0.30 s of delay, an exact-serial performance measurement, a matched comparison of TRIAD’s selector modes, and a robot-to-robot test setup in which a second Gen3 presents the object: the full two-arm handover completes in simulation, and on the two physical arms nine logged runs reached the capture state and closed Robot A’s gripper on the object presented by Robot B, the objective of those sessions. With the published sources, Robot A alone has since executed the four reference scenarios on the physical arm through the closure of the gripper, and the two arms have run the repository’s two-robot procedure together to the same point, with both arms’ joints in the controller’s log. All figures and numbers regenerate from the repository, the hardware quantities from the archived joint records of those runs.

## I. INTRODUCTION

A receiver can settle its three decisions in sequence: an event time is fixed before it is known that any grasp realises it, and a grasp is fixed before it is known that a collision-free route reaches it and that the arm can still retreat with the object in hand. TRIAD represents the decision as a single finite object

$$\xi = (\tau, g, r), \quad (1)$$

the presentation event time, the receiver grasp and the transit route, and chooses it by exhaustive evaluation of a bounded bank of complete plans, each certified from reach to retreat before any cost is formed (Fig. 1).

### Contributions.

- 1) An instantiation of predictive receiving as finite complete-plan selection, built on earliest-feasible rendezvous [1], [2] and a grasp funnel of the kind of [6]: a bank  $\mathcal{T} \times \mathcal{G} \times \mathcal{R}$  generated from one frozen state, copied-state certification of every plan through retreat, a seven-term objective placed on a common search-to-completion basis, and a timing admission rule applied at selection time (Section IV).
- 2) An implementation on `mc_rtc` in which the search runs on a background worker while the control thread keeps its 1 kHz cycle, with one commitment and no replanning, and with every candidate, check and cost written to inspectable records.
- 3) Two further selector modes on the same platform: a receding mode that re-certifies against a robot-independent giver, and a supervisory mode that adds an earliest-feasible rendezvous solver and a directional velocity-authority filter, compared against matched baselines, where the plain predictive selector completes the most scenarios (Section IV-H).
- 4) A robot-to-robot test setup, a second Gen3 acting as the giver inside the same controller, with which the receiver can be exercised repeatably without a human, in simulation and on the two physical arms (Section IV-I).
- 5) Evidence from the repository’s own records: the feasibility funnel, the cost structure of the certified plans, the latency sweep, the computation cost, the mode comparison, the two-arm simulation and the hardware status (Section V).

TRIAD builds on well-established components, which Section II names; what it adds is that the complete plan is explicit, certified as a whole, and auditable after the fact.

## II. RELATED WORK

**Interception timing.** Selecting the rendezvous instant as the earliest time at which the arm can reach the predicted object pose goes back to Croft, Fenton and Benhabib [1] and Hujic et al. [2]. Menon et al. [3] take the first forward time sample at which a bounded-velocity time of flight plus grasp time fits before the object arrives; Salehian et al. [4] take the first time at which the object enters the reachable set; Islam et al. [5] plan for the pose projected forward by a fixed computation bound and replan under a commit cutoff. TRIAD’s admission rule (Section IV-E) is a discretised form of this family; what

control thread, 1 kHz (mc\_rtc FSM + QP)

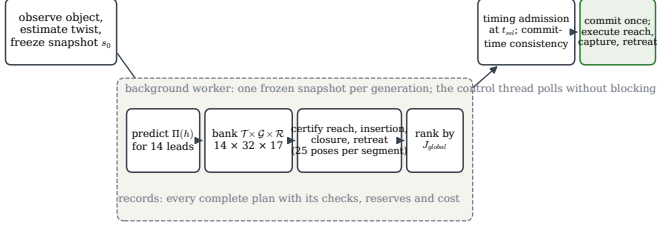


Fig. 1. TRIAD in one cycle of decision. The control thread observes the object and freezes a snapshot; a background worker predicts the presentation pose at each event lead, generates the bank, certifies every complete plan and ranks the survivors; the control thread re-applies timing admission with its own clock, checks prediction consistency, commits once and executes.

differs is that timing is a hard admission gate applied after certification, not a term traded against the objective.

**Grasp selection for moving objects.** Akinola et al. [6] rank a fixed grasp database by precomputed reachability and motion awareness before exact inverse kinematics; Yang et al. [7] maintain a temporally consistent grasp set and resample when it is depleted. TRIAD’s grasp bank is a cheap-to-generate, exactly certified funnel of that kind; its certification of each candidate covers insertion, closure and retreat as well as the reach.

**Handover control.** Yang et al. [8] solve the receiver’s motion with model predictive control over a goal set that couples grasp choice and motion; Oelerich et al. [9] make progress along a path explicit in a path-following MPC; Djeha et al. [10] express handover constraints inside a task-space quadratic programme, which is also TRIAD’s execution layer. TRIAD instead materialises the complete pre-contact plan as one finite object that can be enumerated and inspected. Ortenzi et al. [11] review the field; Mason and MacKenzie [12] measured the human–human transfer that receivers imitate.

### III. PLATFORM AND PROBLEM

The receiver is a 7-DoF Kinova Gen3 with a Robotiq 2F-85 gripper, controlled by an mc\_rtc finite-state controller at 1 kHz. The object is a rigid two-handed body of 0.346 kg: a central sensor cylinder (radius 17 mm, length 36.4 mm) with two handles of radius 11.25 mm and length 137.4 mm centred at  $\pm 86.9$  mm along the object axis; the robot receives the blue handle while the giver holds the grey one. The object pose is provided by a simulated giver that moves the object at constant velocity and stops with a prescribed profile; the transfer force is a virtual spring. Let  $t_0$  be the epoch at which the planning state  $s_0$  (robot configuration, object estimate, model constants) is frozen, and let  $h$  be an event lead measured from  $t_0$ , so that  $t_{\text{event}} = t_0 + h$ . The task is to choose  $\xi$  such that the arm reaches the predicted presentation pose in time, closes the gripper on the handle, and retreats with the object, while the whole plan is known before the arm moves.

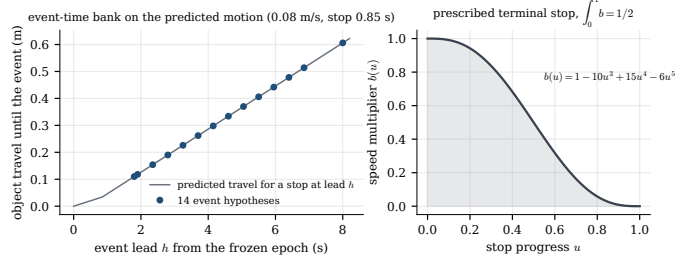


Fig. 2. Event-time bank on the predicted motion. Left: the object travel until the event for each of the 14 leads, on the lateral-low scenario (0.08 m/s). Right: the prescribed terminal stop profile applied at every event hypothesis.

## IV. METHOD

### A. Object estimation and event prediction

A delayed measurement of age  $a$  is propagated to the estimation time with the filtered linear and angular velocity estimates,  $\hat{p}(t) = p_{\text{meas}}(t - a) + a\hat{v}(t)$  and  $\hat{R}(t) = \text{Exp}(a\hat{\omega}(t)) R_{\text{meas}}(t - a)$ ; the velocity filter has a time constant of 0.10 s. For an event lead  $h$  the predicted presentation pose applies a prescribed terminal stop of duration  $D = 0.85$  s with the quintic multiplier  $b(u) = 1 - 10u^3 + 15u^4 - 6u^5$ , whose integral over the stop is one half, so that

$$\Pi(h) = \text{Prop}({}^W T_O(t_0), h - \frac{1}{2} \min(h, D), \hat{v}, \hat{\omega}). \quad (2)$$

The predictor is deterministic: no covariance is propagated and no learned model is used (Fig. 2).

### B. The finite bank

**Event times.**  $\mathcal{T}$  holds 14 leads, 1.80, 1.90, 2.35, 2.80, 3.25, 3.70, 4.15, 4.60, 5.05, 5.50, 5.95, 6.40, 6.85 and 8.00 s: a configured cap of 15 hypotheses with lead step 0.45 s, clamped to the admissible window and deduplicated.

**Grasps.** Let  $\mathbf{z}_H$  be the unit axis of the blue handle and  $\mathbf{p}_H$  its centre, both in the frozen object pose, and let  $\mathbf{b}$  be the direction from the handle centre to the current gripper mouth, projected orthogonally to  $\mathbf{z}_H$  and normalised (“candidate zero” is the direct object-to-mouth direction, so the bank is robot-relative). For an angle  $\theta_i = 2\pi i/16$ ,  $i = 0, \dots, 15$ , and a sign  $\sigma \in \{+1, -1\}$ , the mouth frame  $M$  is built as

$$\begin{aligned} \mathbf{z}_M &= \sigma \mathbf{z}_H, & \mathbf{u}_i &= \text{Rot}(\mathbf{z}_M, \theta_i) \mathbf{b}, \\ \mathbf{y}_M &= \text{norm}(\mathbf{u}_i - \mathbf{z}_M(\mathbf{z}_M \cdot \mathbf{u}_i)), & \mathbf{x}_M &= \mathbf{y}_M \times \mathbf{z}_M, \end{aligned} \quad (3)$$

so that the gripper closes across the handle with its mouth axis  $\mathbf{y}_M$  pointing outward from the handle. The capture point sits a depth of 6 mm in front of the handle centre along  $\mathbf{y}_M$ , the standoff 120 mm further out along the same ray, and the retreat pose 180 mm out: insertion and retreat follow one certified outward ray, so no world direction (above, below, lateral) is preferred by construction. The 16 angles for each sign give the 32 grasps of Fig. 3; they are named by sign and angle, e.g. axisP\_side\_45deg.

**Routes.** For a grasp with standoff pose  $\mathbf{p}_S$  and the current mouth position  $\mathbf{p}_0$ , the chord  $\mathbf{c} = \mathbf{p}_S - \mathbf{p}_0$  defines a plane orthogonal to it with basis  $\mathbf{e}_1$  (world up projected orthogonally

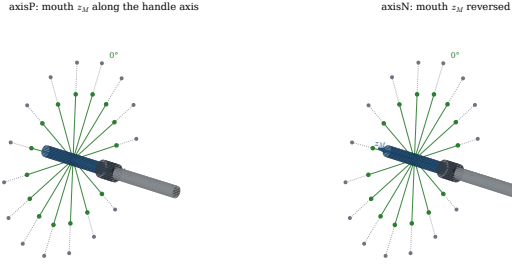


Fig. 3. The grasp bank, drawn from the construction rule of the controller, with the object orientation of the reference scenarios and an illustrative mouth position. Sixteen angles about the blue-handle axis (blue) for each of the two mouth conventions; green rays run from the standoff pose (0.120 m) to the capture point, dotted rays to the retreat pose (0.180 m). Angle 0 is the current mouth direction.

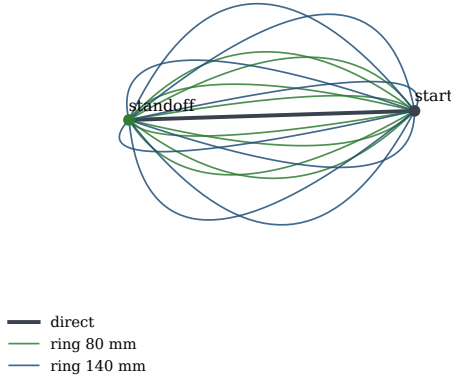


Fig. 4. The route bank for one grasp: the direct start–standoff chord and eight directions at each of two apex offsets in the plane orthogonal to the chord, 17 routes.

to  $\mathbf{c}$ ) and  $\mathbf{e}_2 = \hat{\mathbf{c}} \times \mathbf{e}_1$ . The route bank  $\mathcal{R}$  holds the direct chord and, for each apex offset  $\rho \in \{0.08, 0.14\}$  m and each of  $K = 8$  directions  $\alpha_k = 2\pi k/K$ , a smooth world-space curve through the apex  $\mathbf{p}_0 + \frac{1}{2}\mathbf{c} + \rho(\cos \alpha_k \mathbf{e}_1 + \sin \alpha_k \mathbf{e}_2)$ : 17 routes per grasp (Fig. 4). Routes are generators, not fixed world paths; the ring is sampled symmetrically about the chord and no direction is labelled or preferred. The bank is therefore  $|\mathcal{T}| \cdot |\mathcal{G}| \cdot |\mathcal{R}| = 14 \times 32 \times 17 = 7616$  complete candidates before pruning.

### C. Copied-state certification

Each complete plan is previewed on a copy of the robot state by weighted damped least-squares differential inverse kinematics. With the geometric Jacobian  $J$ , the task twist  $\mathbf{v}$  and a diagonal joint-mobility matrix  $M$  that decreases towards

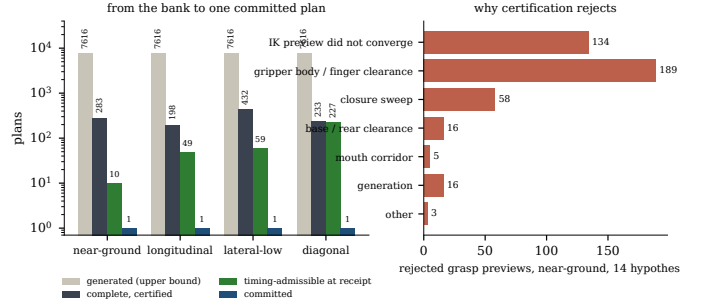


Fig. 5. From the bank to one committed plan, from the repository's records. Left: the four reference scenarios, log scale: the 7616 generated candidates, the complete plans that pass every certification stage, the plans still timing-admissible when the worker's result reached the control thread, and the single commitment. Right: the stage at which rejected grasp previews of the near-ground scenario were stopped, over all 14 event hypotheses.

joint limits (floor 0.02),

$$J^\# = MJ^\top (JM J^\top + \lambda^2 I)^{-1},$$

$$\dot{\mathbf{q}} = J^\# \mathbf{v} + (I - J^\# J)(\dot{\mathbf{q}}_{\text{post}} + \dot{\mathbf{q}}_{\text{lim}}), \quad (4)$$

with damping  $\lambda = 0.040$ , integration step 0.020 s, directional joint-speed clipping and a 0.98 fraction-to-boundary scale before integration; each segment has a finite iteration budget (700). The preview covers, in order, the transit route to the standoff, the insertion into the capture corridor, a closure sweep of the gripper (120 aperture samples), the bilateral-contact conditions, and the carried-object retreat. A commanded swept segment is evaluated at 25 poses. The gripper is modelled by spherical proxies (finger tips, finger bodies, knuckles, pads, base rear) against the object cylinders and the ground plane; a proxy pair  $k$  at sampled pose  $j$  passes if its signed distance satisfies  $d_{k,j} - m_k \geq 0$  for the phase-specific margin  $m_k$ , with hard and soft clearance references of 0.020 and 0.080 m on reach and retreat, joint limits, corridor alignment and terminal tolerances checked separately. A plan enters the hard-feasible set  $\mathcal{F}_h(s_0)$  only when every check has passed, and the cost-valid set  $\mathcal{F}_J(s_0)$  when in addition its objective is finite. The record of every plan, passed or rejected, carries the stage and the reason (Fig. 5).

### D. The objective

For each cost-valid plan the motion objective is

$$J_{\text{motion}} = w_T T + w_E E + w_L L + w_C C + w_Q Q + w_K K + w_V V, \quad (5)$$

with the terms and weights of Table I.  $T$  is the predicted execution time divided by a reference of 8 s;  $E$  an integrated squared joint-speed proxy divided by 8;  $L$  the transit path length divided by 0.50 m;  $V = \text{clip}(\rho_q, 0, 1)^4$  with  $\rho_q$  the maximum directional joint-velocity utilisation.  $C$ ,  $Q$  and  $K$  are reserve penalties of the form

$$\phi(x; x_s) = \begin{cases} 0 & x \geq x_s \\ -\log(\max(10^{-12}, x/x_s)) & 0 < x < x_s \\ 10^6 & \text{otherwise,} \end{cases} \quad (6)$$

TABLE I  
OBJECTIVE TERMS AND WEIGHTS.

| term | preference                              | weight |
|------|-----------------------------------------|--------|
| $T$  | predicted execution time / 8 s          | 0.4211 |
| $E$  | integrated squared joint speed / 8      | 0.1053 |
| $L$  | transit path length / 0.50 m            | 0.1053 |
| $C$  | clearance reserve (log penalty)         | 0.1579 |
| $Q$  | joint-limit reserve (soft 0.20)         | 0.0842 |
| $K$  | conditioning reserve (soft 0.10)        | 0.0737 |
| $V$  | joint-velocity utilisation <sup>4</sup> | 0.0526 |

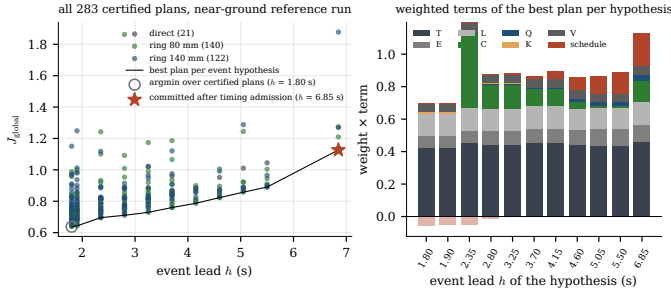


Fig. 6. The objective of the certified plans, near-ground reference run (log in the repository). Left:  $J_{\text{global}}$  of all 283 certified plans against their event lead, by route family; the line joins the best plan of each event hypothesis, the circle is the argmin over all certified plans, the star the plan committed after timing admission with the controller clock (the 3.0 s search had consumed the short leads). Right: the seven weighted terms of the best plan of each hypothesis, stacked, with the schedule term  $J_{\text{global}} - J_{\text{motion}}$  in red (translucent where negative).

applied to the clearance reserve  $(d - d_h)/(d_s - d_h)$  with  $d_h = 0.020$  m and  $d_s = 0.080$  m, to the minimum normalised joint-limit margin with soft reference 0.20, and to the minimum condition index  $\sigma_{\min}(\tilde{J})/\sigma_{\max}(\tilde{J})$  of the Jacobian with its angular rows scaled by a 0.20 m characteristic length, soft reference 0.10. Orientation is enforced by the hard checks; a residual orientation term  $R$  is computed and logged but carries weight zero. The weights are fixed engineering preferences, normalised by their sum.

**Common temporal origin.** Plans attached to different event times are compared on a search-to-completion basis. With  $T_{\text{reach}}$  the plan's predicted reach duration, the waiting time before the reach is  $T_{\text{wait}} = h - T_{\text{reach}}$  and

$$J_{\text{global}} = J_{\text{motion}} + w_T \frac{T_{\text{wait}}}{8 \text{ s}}, \quad (7)$$

which extends the existing time preference back to the epoch rather than adding a second time cost (Fig. 6).

### E. Timing admission and selection

Cost ranking and timing admission are different operations. After the bank has been evaluated, the controller re-applies timing with the actual selection time  $t_{\text{sel}}$ . With  $\text{rem}(\xi) = t_{\text{event}}(\xi) - t_{\text{sel}}$ , the minimum safe commit lead  $L_{\text{safe}} = 1.6$  s and the reach-entry reserve  $L_{\text{reach}} = 0.050$  s, a plan is admissible only if

$$\text{rem}(\xi) + \varepsilon \geq L_{\text{safe}} \quad \text{and} \quad T_{\text{reach}} + L_{\text{reach}} \leq \text{rem}(\xi) + \varepsilon, \quad (8)$$

with  $\varepsilon = 10^{-12}$ . The decision is

$$\xi^* = \arg \min_{\xi \in \mathcal{F}_{\text{timing}}(s_0, t_{\text{sel}})} J_{\text{global}}(\xi; s_0), \quad (9)$$

exhaustive over the bank. Ties within  $10^{-9}$  are broken deterministically by search-to-completion duration, event time, reach duration, larger clearance and candidate name. A plan can have the best objective and still be rejected because too little time remains; Fig. 6 shows exactly this on the near-ground records.

### F. Commitment and execution

Before execution the winner's event pose is re-predicted; commitment requires  $\|p_c - p_f\| \leq 0.015$  m and  $\|\text{Log}(R_c R_f^T)\| \leq 0.12$  rad between the frozen and refreshed predictions. An expired or inconsistent winner is rejected without selecting a replacement. After commitment the plan is executed by the mc\_rtc task/QP layer, which tracks the committed task-space reference (transform task with stiffness 58 and weight 6600, far speed 0.38 m/s, near speed 0.16 m/s, clearance-driven slowdown) together with posture and gripper targets; it does not choose  $\tau$ ,  $g$  or  $r$ , and there is no reselection, retiming or replanning after commitment. Acquisition closes the gripper on bilateral contact and requires the object to be at rest (4 mm/s).

### G. Background worker

The bank search is expensive, so the control thread hands the frozen snapshot to one background worker and continues its cycle; the worker returns its records, and the control thread applies timing admission and the consistency check to them. Worker-result polling never blocks the control cycle. On the reference machine the worker wall time for the near-ground scenario is 2.9 s, against a control-thread maximum of 40.4 ms per cycle, 5.7 ms without logging.

### H. Receding and supervisory modes

Two further modes run on the same platform and share the acquisition interface. The *receding mode* replaces the single frozen search by a supervisor that re-certifies the active plan every cycle, cancels and restarts the search when the object estimate changes, and commits at a terminal boundary; it uses a robot-independent scripted giver (constant twist, 0.40 m travel, quintic stop), so that the event time is a property of the world and not of the robot's commitment. The *supervisory mode* adds a grasp family  $g = (\sigma, \theta, s)$  of 530 hypotheses including an axial offset  $s$  along the handle, a wrist-reachability prune with zero false negatives measured at three offsets, a sampled earliest-feasible rendezvous solver of the Croft-Hujić type whose reference is a Hermite patch between the current and the meeting pose, and a directional velocity-authority index  $\kappa$  computed from the QP's joint-velocity box: the ratio of the velocity the arm can realise along the demanded direction to the velocity the plan demands. Four selectors are defined on it: reactive tracking (B0), predictive rendezvous (B1), predictive with  $\kappa$  as a tie-break (B2), and predictive with  $\kappa \geq 1$  as a hard filter (FULL).

TABLE II  
REFERENCE SCENARIOS: THE COMMITTED PLAN AND ITS COMPLETION  
(SIMULATION).

| scenario     | $h^*$ (s) | route      | $J_{\text{global}}$ | compl. pred./act. (s) |
|--------------|-----------|------------|---------------------|-----------------------|
| near-ground  | 6.850     | ring 80 mm | 1.127               | 10.26 / 8.41          |
| longitudinal | 5.500     | ring 80 mm | 0.807               | 9.11 / 7.76           |
| lateral-low  | 8.000     | ring 80 mm | 0.896               | 9.48 / 8.16           |
| diagonal     | 4.600     | ring 80 mm | 0.699               | 9.00 / 7.63           |

### I. Robot-to-robot coordination

To exercise the receiver without a human, a second Kinova Gen3 (Robot B, base at  $[1.35, 0, 0]$  m, yaw  $\pi$ ) presents the object while Robot A receives. Robot B is coordinated by a giver module inside the same controller: it prepositions to the scenario start, settles, waits for Robot A's object observation to begin, executes the presentation along a fixed world-frame trajectory (0.08 m/s, from  $[0.92, 0, 0.55]$  to  $[0.55, 0, 0.55]$  m, 0.85 s deceleration), settles at the terminal gate and holds; the object is coupled to Robot B's tool until Robot A acquires it. Robot A observes that carried object and, because Robot B's endpoint and arrival time are fixed by its scenario, evaluates the complete grasp  $\times$  route bank once at that single event instead of over an event ladder; the rest of Robot A's FSM is unchanged. An alternative runs Robot B from a second computer with no communication, from a fixed start and a single trigger. On hardware the two arms are driven through per-robot Kortex bridges.

## V. EVALUATION

### A. Reference scenarios

Four moving-object scenarios (Table II) exercise the complete chain. In every one the bank yields a committed plan and the handover completes in simulation; the predicted completion time is conservative by 1.3 to 1.9 s. These are reference runs of the finite-plan mode (the logs are in the repository), not a comparison against another method.

### B. The funnel

Fig. 5 shows what the bank becomes: of 7616 generated candidates, 283, 198, 432 and 233 complete plans pass certification in the four scenarios; of those, 10, 48, 53 and 227 are still timing-admissible when the worker's result reaches the control thread; one is committed. Rejections concentrate in the whole-arm checks: in the near-ground scenario 134 grasp previews stop because the IK preview does not converge, 189 on gripper body or finger clearance, 58 in the closure sweep. Reach alone is a weak filter; the closure and retreat stages do most of the work.

### C. Cost structure

Fig. 6 shows every certified plan of the near-ground reference run against its event lead, and the seven weighted terms of the best plan of each event hypothesis. The argmin over all certified plans lies at the shortest lead, 1.8 s ( $J_{\text{global}}$  0.638); the plan committed after timing admission lies at 6.85 s (1.127), because

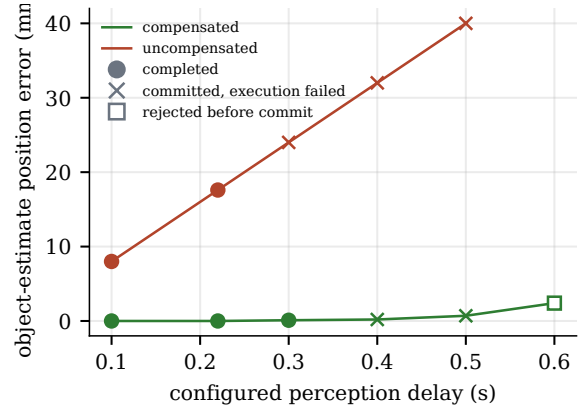


Fig. 7. Perception-latency sweep from the sweep records: object-estimate error against configured delay for the compensated and uncompensated modes, with the run outcome as marker.

the 3.0 s search had consumed the short leads. Across the hypotheses the execution-time term is the largest contribution and nearly constant, the reserve terms separate plans at one event time, and the schedule term grows with the lead.

### D. Perception latency

A timestamped delay is injected into the object-pose measurement and either propagated by its measured age (compensated) or treated as current (Fig. 7). Both modes complete at 0.10 and 0.22 s. At 0.30 s only the compensated mode completes, four of four repeats against zero of four. From 0.40 s both modes commit and fail in execution; at 0.60 s the compensated mode is rejected before commitment by the freshness gate and the uncompensated mode cannot classify the object (displacement 0.0241 m under the 0.0250 m moving threshold while the estimated speed exceeds the static threshold). Compensation keeps the estimate error below 1 mm where the uncompensated estimate drifts by 40 mm at 0.5 s, and extends the usable delay from 0.22 s to 0.30 s.

### E. Computation

Restructuring the serial search reduced the state-scoped wall time over the four scenarios from 4.349 to 3.972 s, 3.315 to 3.072 s, 3.393 to 3.162 s and 2.797 to 2.620 s (6.3 to 8.7 %), with a collision oracle of 8 168 732 pairwise comparisons and zero mismatches against the reference. With the background worker the control-thread cycle stays at or below 40.4 ms while the search runs (5.7 ms excluding logging).

### F. Selector modes

The four supervisory-mode selectors were run over the four scenarios with three near-deterministic repeats each, on an identical plant, an identical grasp pool and one evaluator (Fig. 8). Predictive rendezvous completes 11 of 12 against 5 of 12 for reactive tracking: it meets the grasp pose 1.5 to 3.1 mm from the target and up to 1.6 s before the giver stops. The capability tie-break neither helps nor hurts (10 of 12). The hard authority filter completes 6 of 12: used as a hard constraint it makes selection depend on whether the

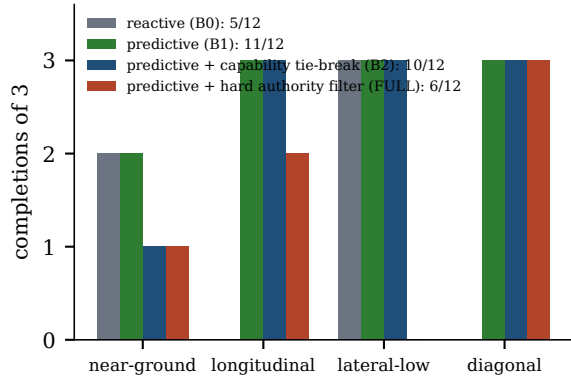


Fig. 8. Completions per selector and scenario, three repeats each, from the campaign records.

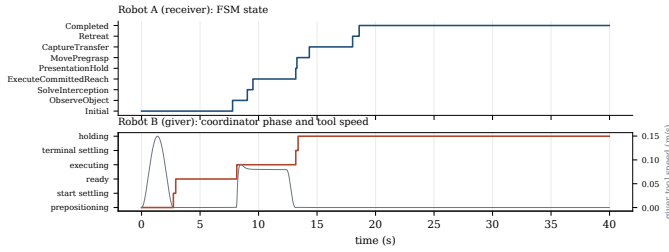


Fig. 9. Two-arm handover in simulation, from the run log: receiver state and giver phase with the giver's tool speed.

arm starts at rest or moving and produces adopt/abort cycling (140 aborts over the three longitudinal repeats, up to 49 in one run). Directional authority carries information,  $\kappa < 1$  in 35 to 43 % of preview-feasible encounters and a median Spearman correlation of  $-0.83$  against the condition index, but the recommended configuration is predictive rendezvous with  $\kappa$  as a diagnostic, not a filter. A first campaign had reported FULL at 12 of 12; it came from a rollout defect on replans from a moving arm that inflated the demand signal the filter consumes, and both campaigns are archived.

### G. Robot-to-robot handover

With Robot B as the giver the full two-arm handover completes in simulation (Fig. 9, Fig. 10): Robot B settles from 2.7 s, is ready at 2.9 s, executes from 8.1 s in sync with Robot A's observation and holds from 13.4 s; Robot A observes from 7.8 s, commits at 9.5 s to Robot B's endpoint (lead 4.15 s, search 0.49 s), captures at 14.3 s, retreats at 18.0 s with the object and completes at 18.6 s, with an execution of 8.2 s against 9.6 s predicted. The object pose Robot A observes, plans and retreats with and the pose Robot B carries differ by at most 0.5 mm over the run (the timeline in the repository carries both). The single-robot scenarios complete on the same build, and so does a run with the second robot loaded and the giver disabled.

### H. The two physical arms

Both arms have been driven together on hardware through per-robot Kortex bridges. Fig. 11 shows the laboratory setup

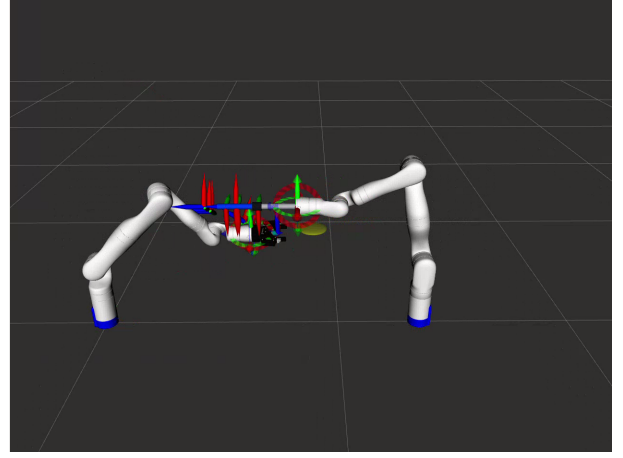


Fig. 10. The two-arm simulation: Robot B (right) presenting the object to Robot A (left).

on video: Robot B presents the object, a bottle held in its tool, and Robot A approaches with its Robotiq gripper and closes it on the bottle neck. The controller logs of those sessions contain the second arm's motor currents, which only a physical robot produces; they record Robot B alone executing its presentation through all six coordinator phases while Robot A held, and nine combined runs in which Robot A observed, planned, reached, held, approached, entered the capture state with the physical gripper bridge enabled and closed the gripper on the object. Reach and close was the objective of those sessions: the object was taped to Robot B's tool, so no load transfer and no retreat were attempted by design. After the closure each run ended in the fail-safe hold inside the capture state, because the closure check against the virtual object model reported pad contact (the controller has no physical contact or force signal), as expected in that setup. The logs are not synchronised with the video, so the video and the logs are reported as separate evidence. Robot A's gripper has been commissioned on hardware with a measured closure calibration, and the receiver's hardware switches and that calibration are collected in one configuration overlay in the repository. The transfer force source was virtual in every hardware run, and the integrated giver coordinator reported in Section IV-I has been exercised end-to-end in simulation; on hardware Robot B alone ran with that coordinator, and the combined runs used the same coordinator inside the laboratory's sandbox controller (in four early runs of the last session the Kortex driver instead interpolated Robot B's joints to a fixed target on the presentation time law; that override is kept in the driver source, off by default).

On 25 September 2026 the procedure was rerun from the published repository state. Robot A alone first executed the four reference scenarios of Section V against the virtual object on the physical arm: readiness posture, observation, one committed plan, certified reach, presentation hold, pregrasp and closure of the physical gripper at the planned capture pose in every scenario (reach targets from  $[0.163, 0.195, 0.166]$  to

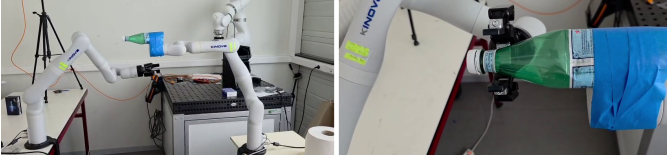


Fig. 11. The two physical arms in the laboratory, from the video record. Left: Robot B (right arm) presents the bottle while Robot A (left arm, Robotiq gripper) approaches. Right: Robot A's gripper closes around the bottle neck while Robot B still supports it.

$[0.475, 0.096, 0.556]$  m, clearances of 54 to 82 mm, measured joints within 0.03 rad of the commanded ones), ending in the fail-safe hold at the closure check with nothing between the fingers. One start lost its commitment to a 4.3 s search, against 2.7 to 3.2 s otherwise, and held without moving: timing admission refuses a late plan rather than extrapolating the event. Both arms then ran the two-robot procedure: Robot B alone through its six phases to the hold with 0.2 mm endpoint error, then the handover, in which Robot A committed once at Robot B's fixed endpoint, reached to  $[0.463, 0.108, 0.486]$  m with 80 mm clearance in step with the presentation and closed the gripper at the planned object pose (Robot B moving up to  $52^\circ$ , Robot A up to  $106^\circ$  on its wrist joints plus one full revolution of its continuous third joint during the readiness move, when its initial reading was  $-\pi$  rather than  $+\pi$ ; tracking within 0.02 rad). No object was mounted, so the run ended, as in July, at the closure check. The joint records of these runs and the script that recomputes the quoted quantities are in the repository; contact, load transfer and retreat remain to be exercised with a physical object and a contact signal.

## VI. DISCUSSION

Three properties of the method follow from its own records. First, timing dominates: in three of the four reference runs the search latency moves the committed event time away from the argmin over certified plans, and the objective of the committed plan is carried mostly by its time terms. TRIAD makes that visible instead of hiding it inside a continuously re-solved problem. Second, certification through retreat matters: in the near-ground record most rejections occur in the clearance and closure checks, not at reach (Fig. 5), so a reach-only selector would commit to plans that cannot finish. Third, the simulated giver of the default configuration follows the committed presentation time after commitment, so in that configuration the event time is a schedule the robot sets; the receding mode with its independent giver removes this, at the cost of moving-epoch searches longer than the giver's constant-velocity window and a prediction validity of 0.10 to 0.65 s while the object moves. The limits are equally clear: acquisition requires the object to stop, so no mode acquires a moving object; the evidence is simulation with virtual force; the bank is an engineering discretisation with no resolution study; the objective weights are fixed preferences and off-line replay shows a lexicographic rule (completion, then clearance, then effort) captures their useful part. What is reusable is the

formulation, the plant, the certification stage, the evaluation tooling, and a two-robot setup on which the next hardware run can be made repeatably.

## VII. REPRODUCIBILITY

The controller, configuration, evidence records, the two-robot setup, the unit tests, the documentation-claim guards and the script that regenerates every figure of this paper from the records are in the TRIAD repository (tag triad-2026-09-24). The two-arm simulation runs from a build tree with one script; the hardware procedure is documented step by step.

## VIII. CONCLUSION

TRIAD chooses event time, grasp and route together, certifies each complete plan through retreat, and commits once. Its records show the shape of the decision: a bank of thousands, a few hundred certified plans, a few dozen admissible ones, one commitment dominated by timing. The platform includes a second arm as the giver: the full handover completes in simulation, and on the two physical arms nine logged runs reached the capture state and closed Robot A's gripper on the object presented by Robot B, which was the objective of those sessions; the next step is the integrated coordinator on the two physical arms with synchronised logs.

## REFERENCES

- [1] E. A. Croft, R. G. Fenton, and B. Benhabib, "Optimal rendezvous-point selection for robotic interception of moving objects," *IEEE Trans. Syst., Man, Cybern. B*, vol. 28, no. 2, pp. 192–204, 1998.
- [2] D. Hujić, E. A. Croft, G. Zak, R. G. Fenton, J. K. Mills, and B. Benhabib, "The robotic interception of moving objects in industrial settings: strategy development and experiment," *IEEE/ASME Trans. Mechatronics*, vol. 3, no. 3, pp. 225–239, 1998.
- [3] A. Menon, B. Cohen, and M. Likhachev, "Motion planning for smooth pickup of moving objects," in *Proc. IEEE ICRA*, 2014, pp. 453–460.
- [4] S. S. Mirrazavi Salehian, N. Figueroa, and A. Billard, "Coordinated multi-arm motion planning: reaching for moving objects in the face of uncertainty," in *Proc. Robotics: Science and Systems*, 2016.
- [5] F. Islam, O. Salzman, A. Agarwal, and M. Likhachev, "Provably constant-time planning and replanning for real-time grasping objects off a conveyor belt," in *Proc. Robotics: Science and Systems*, 2020.
- [6] I. Akinola, J. Xu, S. Song, and P. K. Allen, "Dynamic grasping with reachability and motion awareness," in *Proc. IEEE/RSJ IROS*, 2021.
- [7] W. Yang, C. Paxton, A. Mousavian, Y.-W. Chao, M. Cakmak, and D. Fox, "Reactive human-to-robot handovers of arbitrary objects," in *Proc. IEEE ICRA*, 2021.
- [8] W. Yang, B. Sundaralingam, C. Paxton, I. Akinola, Y.-W. Chao, M. Cakmak, and D. Fox, "Model predictive control for fluid human-to-robot handovers," in *Proc. IEEE ICRA*, 2022.
- [9] T. Oelerich, C. Hartl-Nesic, and A. Kugi, "Model predictive trajectory planning for human-robot handovers," arXiv:2404.07505, 2024.
- [10] M. Djeha, A. Dallard, A. Zermane, P. Gergondet, and A. Kheddar, "Human-robot handovers using task-space quadratic programming," in *Proc. IEEE RO-MAN*, 2022.
- [11] V. Ortenzi, A. Cosgun, T. Pardi, W. P. Chan, E. Croft, and D. Kulić, "Object handovers: a review for robotics," *IEEE Trans. Robotics*, vol. 37, no. 6, 2021.
- [12] A. H. Mason and C. L. MacKenzie, "Grip forces when passing an object to a partner," *Exp. Brain Res.*, vol. 163, 2005.